

AY: 2025-2026
Course: PPOM 1
Class of 2029
Week 13, Lecture 121

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DPR Lecture: History and Physical exam of the Spine

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Session Objectives

- Demonstrate the components of the history taking and physical examination of the Spine
- Describe subjective and objective findings relating to the examination of the spine
- Distinguish Normal vs. Abnormal examination findings
- Recall the appropriate way to document common normal and abnormal findings of the history and physical Examination
- Incorporation of OMM in a Case Study

RED WOLVES and BEARS

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The Health History

- Common or Concerning Symptoms
 - Joint pain: articular or extra-articular, acute or chronic, inflammatory or noninflammatory, localized or diffuse, monoarticular or polyarticular
 - Joint pain: associated constitutional symptoms and systemic manifestations from other organ systems
 - Fever, chills, rash, weight loss, weakness
 - Neck pain
 - Low back pain

History- Low Back Pain

- Is the pain on the midline or off the midline
- Is there radiation into the buttock or lower extremity?
- Associated numbness or paresthesia?
- Leg pain that resolves with rest and/or lumbar forward flexion
- Is there an associated bowel or bladder dysfunction?
- “Red Flags” for serious underlying systemic disease

History Taking- Back Pain

- Assessed decreased or limited movement
- Timing
 - Acute vs. Chronic
 - What aggravates or relieves the pain
- Systemic Features
 - Fever? Chills? Rash? Fatigue? Anorexia? Weight loss? Weakness?

“Red Flags” for Low Back Pain

- Age older than 50 years or less than 20 years
- History of cancer
- Unexplained weight loss, fever, or decline in general health
- Pain lasting more than 1 month or not responding to treatment
- Pain at night or increased by rest

“Red Flags” for Low Back Pain

- History of IVDA, addiction, or immunosuppression
- Presence of active infection or HIV
- Long-term steroid therapy
- Saddle anesthesia, bladder or bowel incontinence
- Neurologic symptoms or progressive neurologic deficit

Common Review of System

- Constitutional: Fever, chills, or fatigue
- Derm: Rash or sores
- HEENT: Head injury, blurry vision, ear pain or sinus pain
- Neck: Neck pain
- Respiratory: SOB or difficulty breathing
- CV: Chest pain or palpitations
- GI: N/V/D/C; Abdominal pain

Common Review of System

- Peripheral Vascular: No varicose veins
- MSK: Any other joint pain, muscle pain/weakness
- Neuro: Numbness / Tingling / Weakness / Saddle Anesthesia / loss of bowel and bladder control
- Endocrine: Excessive thirst / hunger / urination
- GU: F / U / D; Dysuria; Discharge
- Psychiatric: Depression or anxiety

Physical Examination- Spine

- Should be systematic
 - Inspection
 - Palpation
 - Range of Motion
 - Active
 - Passive
 - Strength testing
 - Special testing

Physical Examination- Spine

- Assessing the 4 signs of Inflammation
 - Swelling
 - Warmth
 - Redness
 - Pain or tenderness
 - Diffuse
 - Focal

Examination of the Cervical Spine

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Cervical Spine- INSPECTION

- Should reveal sufficient cervical lordosis
- Inspect the patient's posture
- Positioning of the head
- Abnormal rotation or side bending
- The muscle bulk of the trapezius, deltoid, and SCM muscles should be symmetrical

Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Examination of the Cervical Spine

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Cervical Spine- PALPATION

- Sitting or standing position
- Spinous processes
 - Note pain, swelling, step-offs
 - Any cervical spine with spinous processes that show either a step-off or a significant difference in interspinous distance needs radiographs
- Laterally and palpate the musculature in the posterior neck
- Suboccipital fossae noting tenderness and somatic dysfunction
- Greater occipital nerves may be tender in patients with chronic headaches

Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Cervical Spine- Range of Motion

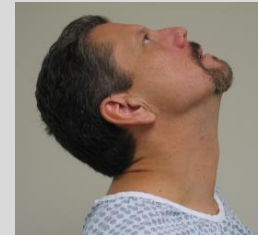
- Neck is the most mobile of the spine
 - Flexion- “Bring your chin to your chest”
 - Extension- “Look up at the ceiling”
 - Rotation- “Look over one shoulder, and then the other”
 - Lateral bending- “Bring your ear to your shoulder”
- Cervical Flexion and extension → OA joint
- Cervical Rotation → AA joint
- Lateral bending → C2-C7

Sources:

1. Bates’ Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Cervical Spine- Range of Motion Testing

- Active Range of Motion
 - Instructions to Patient
 - Extension → “Look up at the ceiling”
 - Flexion → “Bring your chin to your chest”



Cervical Spine- Range of Motion Testing

- Active Range of Motion
 - Instructions to Patient
 - Rotation → “Look over one shoulder, then the other”
 - Lateral bending → “Bring your ear to your shoulder, then to the other”
(Make sure they don’t raise the shoulder)



Cervical Spine- Range of Motion

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- Rotation
- Flexion
- Extension
- Lateral bending
(Side-bending)

Cervical Spine- Range of Motion

- Normal Range of motion for the cervical spine
 - Flexion 60-90 degrees
 - Extension 70-90 degrees
 - Side bending 20-45 degrees
 - Rotation 70-90 degrees
- Note pain reproduction, amplitude of motion and end-feel

Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Cervical Spine- Resisted Range of Motion

- Typically an isometric contraction
 - Painful motion
 - Weakness
 - Or, both
- Limited range of motion
 - Stiffness, pain, overuse, and muscle spasm

Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Cervical Spine Examination

- Neurovascular Examination
 - Muscle Strength
 - Sensation
 - Deep Tendon Reflexes

Muscle Testing: Cervical Spine

Myotome and Dermatome Testing

Nerve Root Level	Sensory Testing	Motor Testing	Reflex Testing
C1-C2	Front of face	Neck flexion	N/A
C3	Lateral face and skull	Lateral flexion	N/a
C4	Supraclavicular	Shoulder shrug	N/A
C5	Lateral shoulder/upper arm	Shoulder abduction	Bicipital (musculocutaneous)
C6	Lateral lower arm and hand (thumb and index finger)	Elbow flexion and wrist extension	Brachialradial (musculocutaneous)
C7	Palmar aspect of hand – middle 3 fingers	Elbow extension and wrist flexion	Triceps (radial)
C8	Medial lower arm and hand	Finger flexion and thumb extension	N/A
T1	Medial elbow and upper arm	Finger abduction	N/A

Source:
<http://www.slideshare.net/DrKKalaivananDrKK/c-spine-injuries-dr-kalaivanan-kanniyan>

Cervical Spine Examination

- Lhermitte's Sign
 - A.K.A. Barber's chair phenomenon
 - While sitting on the exam table and attempting to actively hold their head upright, the patient complains of electrical shock sensations down the spine, arms, or legs
 - Indicates possible Spinal stenosis (Rule out Multiple Sclerosis, ALS, Arnold-Chiari Malformation)

Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Axial Compression Test

- Examination
 - Standing behind
 - Perform test once
 - Examiner places both hands on top of patient's head
 - Exerts a downward force with both hands
- Positive test
 - Pain and/or neurological symptoms of the cervical spine and/or upper extremity are reproduced with the compressive force
- Indicates
 - Cervical Nerve Impingement

Foraminal Distraction Test

- Examination
 - Standing at side of table facing the patient
 - Perform test once
 - Examiner places one hand under the patient's chin and the other under the occiput, then applies an upward traction force on the cervical spine
- Positive test
 - Pain and/or neurological symptoms of the cervical spine and/or upper extremity are reduced with the traction force
- Indicates
 - Cervical Nerve Impingement

Spurling's Test

- Examination
 - Test for foraminal compression on a nerve root
 - Examiner places one hand on the top of the patient's head and the other on patient's shoulder at the affected side
 - Examiner proceeds to side bend and rotate the patient's neck TOWARD the side of their pain
 - Adds slight extension of the patient's neck
 - The examiner exerts a downward force on the patient's spine
- Positive test
 - Pain radiating down the arm to which the head is rotated
- Indicates
 - Cervical radiculopathy

Sources:

1. ICC Guide
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 9

Cervical Spine- Common Conditions

- Soft Tissue Injuries / Conditions
 - Cervical strain, cervical sprain
 - Most Common
 - Soft tissue activity threshold is surpassed
 - Usually self-limited
 - Cervical Somatic dysfunction
 - Headache aka “cephalgia”
 - Stingers / Burners

Source: Stockard, Alan, *Osteopathic Clinical Joint Exam*; American Osteopathic Academy of Sports Medicine; Chapter 1

Cervical Spine- Common Conditions

- Hard Tissue Injuries / Conditions
 - Osteoarthritis
 - Common condition whereby the multiple joints and vertebral bony components develop the classic changes associated with OA
 - Degenerative disc disease
 - Wear and tear phenomena that occur across the disc-vertebrae interface
 - Associated with aging and the inherent loss of water content within the disc itself

Cervical Spine- More Urgent

- Fracture
 - Moderate forces are at play
 - Pain and dysfunction are disproportional to the physical findings
 - Pinpoint tenderness
 - Neurologic finding
- Herniated Disc
- Infection
- Inflammatory Condition
- Malignancy
- Myelopathy
- Thoracic outlet Syndrome
- Vascular Emergency

Source: Stockard, Alan, *Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine*; Chapter 1
Childress MA, Stueck SJ. Neck Pain: Initial Evaluation and Management. Am Fam Physician. 2020 Aug 1;102(3):150-156.
PMID: 32735440.

Cervical Spine- Imaging for trauma

- NEXUS criteria
 - Well-validated clinical decision aid
 - Safely rule out cervical spine injury
 - 5 Criteria- If none are present, the C-Spine can be cleared clinically
 - Neurologic deficit
 - Spinal Tenderness (Midline)
 - Altered Mental Status / Level of Consciousness
 - Intoxication
 - Distraction (painful) Injury

Nexus Criteria in play

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Photo Credit: Pablo Silva

Nexus Criteria in play

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Photo Credit: Pablo Silva

Cervical Spine- Imaging for trauma

- Canadian C-spine Rule
 - Used in Alert and Stable Trauma patient with C-Spine injury is a concern
 - Includes age (over 65 years)
 - Includes mechanism of injury
 - Low risk factors to allow assessment of ROM
 - Testing of neck rotation (45 degrees to left and right)

Thoracic-Lumbar Spine- INSPECTION

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- “I’m inspecting your back”
- Note any abnormalities
 - Scars
 - Asymmetry
 - Thoracic and lumbar curves



Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. ICC Guide

Thoracic-Lumbar Spine- PALPATION

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- On skin
 - Thoracic lumbar junction
 - Thoracic, lumbar spine
- Palpate paraspinal muscles
- Palpate the spinous process of each vertebra with thumbs
- Evaluates Skin-drag
- Performs Erythema test



Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. ICC Guide

Thoracic-Lumbar Spine- Range of Motion

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- Flexion • Extension • Rotation • Side-Bending



Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. ICC Guide

Thoracic-Lumbar Spine- Range of Motion

- **Flexion (Active)**
 - Examiner instructs patient to bend forward and touch toes, noting smoothness and symmetry of movement, the range of motion, and the curve in the lumbar area
- Extension
- Rotation
- Side Bending

Thoracic-Lumbar Spine- Range of Motion

- Flexion (Active)
- **Extension (Active)**
 - Examiner places one hand on the posterior superior-iliac spine, fingers pointed to mid-line to support the patient's back, then instructs patient to bend back as far as possible
- Rotation
- Side Bending

Thoracic-Lumbar Spine- Range of Motion

- Flexion (Active)
- Extension (Active)
- **Rotation (Passive)**
 - Examiner places one hand on patient's iliac crest (to stabilize the pelvis) and the other hand on patient's opposite shoulder.
 - The examiner rotates patient's trunk by pulling the shoulder and then the hip posteriorly
 - Examiner changes his/her hand positions and repeats procedure for the opposite side
- Side Bending

Thoracic-Lumbar Spine- Range of Motion

- Flexion (Active)
- Extension (Active)
- Rotation
- **Side Bending** (Active)
 - Examiner stabilizes patient's pelvis by placing one hand on the patient's iliac crest and then instructs patient to lean as far as possible on the ipsilateral side.
 - Examiner changes his/her hand position and repeats procedure for the opposite side

Lumbar Spine- Advanced Exam

–Deep tendon reflexes

- L4 patellar, no L5, S1 Achilles

–Muscle strength

- Ankle dorsiflexion (L4), great toe dorsiflexion (L5), plantarflexion S1

–Sensory

- Medial foot L4, dorsal foot L5, lateral foot S1

Thoracic-Lumbar Spine Special Tests

- Scapular Elevation
- Scapular Protraction (Winging)
- Adson's Test
- Adam's Forward Bend Test
- Stork Test
- Straight Leg Raise test
- Bragard's Test
- Well (Opposite) Straight Leg Raising
- Hoover Test

Scapular Elevation

- Examination
 - The examiner stands behind the patient and places his or her hands on each acromion.
 - With arms resting at the side, the patient shrugs his or her shoulders against examiner resistance
- Positive Test
 - Weakness or pain against resistance
- Indicates
 - Tendonitis (of trapezius and/or levator scapulae muscles), cervical injury, or cranial nerve damage (Spinal Accessory nerve, CN XI)

Scapular Protraction (Winging)

- Examination
 - The patient pushes against a wall with both hands with the feet farther away from the wall than the shoulder.
- Positive Test
 - Scapular winging, pain, and/or weakness during the maneuvers
- Indicates
 - **Serratus anterior muscle** weakness. Lateral winging can be generated by trapezius and rhomboid paralysis

Scapular Protraction (Winging)

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Source: Bates' Guide to the Physical Examination
and History Taking; 12th Ed. – Chapter 17

Adson's Test

- Examination
 - The examiner locates the radial pulse.
 - The patient rotates their head toward the tested arm and lets the head tilt backwards (extends the neck) while the examiner abducts/extends the arm.
 - The patient is asked to inhale and turn the head toward the affected side
- Positive Test
 - The disappearance or diminution of the pulse or provocation of symptoms
- Indicates
 - Thoracic outlet syndrome, usually related to the scalene musculature

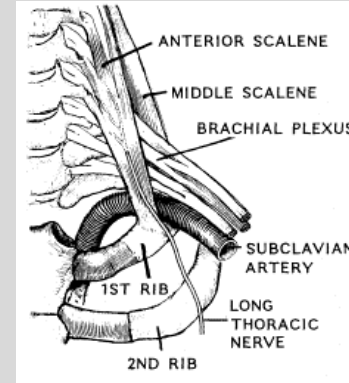
Sources:

1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 5
2. PowerPoint Presented to Class of 2021- *Foundations of Osteopathic Medicine*- OMM Lab #3 – “Osteopathic Evaluation of the Shoulder” presented August 23rd and 24th, 2017

Adson's Test

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Note: Slide reproduced from PowerPoint Presented to Class of 2021-
*Foundations of Osteopathic Medicine- OMM Lab #3 – “Osteopathic
Evaluation of the Shoulder”* presented August 23rd and 24th, 2017

Adam's Forward Bend Test

- Examination
 - Patient stands with feet together
 - Examiner instructs patient to bend spine forward while keeping their feet together, knees straight and their hands hanging at their side
- Positive Test
 - Presence of unilateral rib cage elevation or unilateral lumbar paravertebral muscle prominence
- Indicates
 - The curve of structural scoliosis is more apparent when the patient forward bends

Normal

Abnormal

Stork Test

- Examination
 - Single leg weight-bearing hyperextension test where the patient stands on one leg and then leans backward extending the lower back
- Positive Test
 - The patient experiences pain in the low back, as this stresses the posterior elements of the spine on the ipsilateral side
- Indicates
 - A possible pars defect / stress fracture, spondylolysis, or facet pathology

Straight Leg Raise Test

- Examination
 - Patient lying supine and relaxes his/her leg
 - The leg with knee extended is passively raised off the examination table
 - Bilateral Exam. Uninvolved side first.
- Positive Test
 - A radicular pain emanating from the low back following the sciatic nerve distribution, and usually radiating below the knee on the ipsilateral side
- Indicates
 - Nerve root impingement

Sources:

1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 6
2. PowerPoint Presented to Class of 2021- *Foundations of Osteopathic Medicine*- OMM Lab #11 – “Osteopathic Evaluation of the Hip” presented November 1st and 2nd, 2017

Straight Leg Raise Test

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Note: One can further support nerve root entrapment by Bragard's Test

Source: ICC Guide

Bragard's Test

- Examination
 - By taking the leg to the point just below the level of the pain and adding dorsiflexion of the foot, the L5 portion of the sciatic nerve will be stretched leading to pain
- Positive Test
 - Pain that radiates below the knee
- Indicates
 - Nerve Root Impingement

Well (Opposite) Straight Leg Raising

- Examination
 - Patient raises his/her straight leg on the uninvolved side
- Positive Test
 - Pain is experienced on the side opposite that being raised
- Indicates
 - The test is positive for possible space occupying lesion if pain occurs on the opposite side

Source: ICC Guide

Hoover Test

- Examination
 - With the patient lying supine, the examiner places a hand under each heel.
 - The patient is then asked to lift the affected leg off the table.
 - A normal response is for the patient to push downward with the contralateral heel
- Positive Test
 - No downward force is felt, the patient is not giving enough effort when the patient is raising the involved leg
- Indicates
 - The patient may be malingering, or seeking a secondary gain

Lumbar Spine- Differential Diagnosis

- Lumbar Strain/Sprain
- Herniated nucleus pulposus
- Spinal Stenosis
- Spondylolisthesis
- Spondylolysis
- Spondylosis (degenerative disc disease)
- Compression fracture

Lumbar Spine- Serious Etiologies

- Cauda Equina Syndrome (CES)
- Cancer
- Fracture
- Infection

Case Study

- CC: “Back Pain”
- HPI: 77 y/o male presents to the clinic with chronic upper and lower back pain. Pain has been persisting for about 10 years, around the time he has been diagnosed with Parkinson’s Disease. Patient points to the lower back bilaterally when asked to localized the pain. The pain today is about a 3-4/10 on pain scale, described as a dull achy pain. There is no radiation to pain.

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Pain is worst after prolonged standing and walking, and improves with rest. Workup thus far includes ortho evaluation, epidural injections in the past, and physical therapy with some improvements. He received prior imaging in 2015 of the lumbosacral spine showing moderate degenerative disc narrowing and small central disc herniation, mostly in the lower lumbar regions.

Case Study

- ROS:
- Constitutional: No weight loss or weight gain, no fever / chills, no loss of appetite, no fatigue, + sleeping difficulty
- HEENT: No throat pain, No change in vision or hearing
- Cardio: No chest pain, No palpitations, No dizziness
- Pulm: No shortness of breath, No cough
- GI: No abdominal pain, No nausea, No vomiting, No diarrhea, No constipation, No blood in the stool
- Neurology: No headaches, No numbness or tingling, No urinary or fecal incontinence, No saddle anesthesia

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Allergies: PCN (rash)
- Medications: Selegiline 5mg bid, amlodipine 5mg daily, claritin 10mg as needed
- PMHx: Parkinson's Disease (diagnosed 10 years ago), Hypertension (since age 55)
- PSHx: Appendectomy (age 30)
- FH: Mother, deceased at age 92, HTN; Father, deceased at age 87, CAD
- Social history: Denies history of tobacco or recreational drug use. Denies alcohol use. Retired Accountant. Limited exercise secondary to pain. Diet includes fruits and vegetables.
- HCM: Last colonoscopy 10 years ago was normal. He received the flu vaccine and pneumovax this year. Last physical exam was 3 months ago.

Case Study

- Physical Exam:
 - Vitals: BP- 124/80, HR 74, RR 14, Temp. 97.9, Weight- 165 pounds, Height- 72 inches, BMI- 22.38, Pulse Ox- 98% room air
 - General: Awake, alert, and oriented x 3, No acute distress, pleasant affect
 - Head: Normocephalic, atraumatic
 - Extremity: No clubbing / cyanosis / edema, pulses 2+ bilaterally in both upper and lower extremities

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Physical Exam:
 - Neurology: Normal mental status with hypophonic and tachyphemic speech. Motor strength limited by increased tone bilaterally. Gait: uses walker to ambulate and has pronounced camptocormia

Camptocormia



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Photo taken from: J Neurosurg 112:1283–1288, 2010.DOI: 10.3171/2009.10.JNS09917 which was accessed online: <http://www.neurosurgery-blog.com/archives/tag/camptocormia>

Case Study

- Physical Exam:
 - MSK: Thoracic spine with + head forward and rounder shoulder posture. There is no vertebral spine tenderness. There is paraspinal muscle spasm presents bilaterally. There is increased Thoracolumbar spinal flexion. Range of motion extremely limited In all direction

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Physical Exam:
 - OSE:
 - Sub-occipital hypertonicity, OA E SI Rr, SEVERITY: Moderate, Treated with: Sub-occipital release, BLT indirect, TX RESPONSE: Improved
 - Cervical region with hypertonicity of bilateral upper trapezius, C4, FRS left, Paravertebral muscle hypertonicity bilaterally, SEVERITY: Severe, Treated with: Facilitated Positional Release, Myofascial release, BLT indirect, TX RESPONSE: Improved

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Physical Exam:
 - OSE:
 - Thoracic: Paravertebral muscle hypertonicity on the left, with muscle spasm on the right, SEVERITY: Severe, Treated with: Myofascial Release, BLT indirect, TX RESPONSE: Improved
 - Lumbar: Paravertebral muscle hypertonicity on the left, with muscle spasm on the right greatest at L1-L3, AL1 tenderpoint on the right, SEVERITY: Severe, Treated with: BLT indirect, Counterstrain, TX RESPONSE: Improved

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Physical Exam:
 - OSE:
 - Upper extremity: Fascia restrictions in bilateral shoulders resulting in decreased range of motion, SEVERITY: Moderate, Treated with: Modified Spencer's Technique in supine position with Direct Fascial Release

Case Study

- Assessment:
 - Back pain with multifactorial etiology
 - Degenerative Disc Disease
 - Herniated Disc
 - Spondylosis
 - Multiple Somatic Dysfunctions (Cranial, Cervical, Thoracic, Lumbar)
 - Parkinson's Disease
 - Camptocormia

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Plan:
 - OMM treatment done today in office
 - Proper hydration and healthy nutrients
 - May apply heat as needed
 - Back stretches and ROM exercises advised
 - Continue current Physical Therapy for improvements in balance and gait
 - Tylenol / NSAIDs as needed for pain
 - Will consider getting repeat imaging if symptoms persists or worsens
 - Follow up in 1 week, sooner if symptoms worsens

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study- 2nd Visit (11 days after initial visit)

- At follow up visit, another treatment was given with the following results:

Patient tolerated ÖMM treatment today with great results and when asked how he felt afterwards, he answered, "Better." After treatment, patient was able to fully lift his back to an upright position which is something he has not been able to do prior.

Case Study- 3rd Visit (18 days after initial visit)

- At follow up visit, another treatment was given with the following results:

6/25/18- 74 y/o male presents for follow up visit and repeat omm treatment. Last seen last week and is very happy with the treatment. He noted standing outside and able to stand tall and look at the sky, which is something he was not able to do prior.

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Summary Slide

- Demonstrate the components of the history taking and physical examination of the Spine
- Describe subjective and objective findings relating to the examination of the spine
- Distinguish Normal vs. Abnormal examination findings
- Recall the appropriate way to document common normal and abnormal findings of the history and physical Examination
- Incorporation of OMM in a Case Study

Summary Slide

- Core References:

- Bickley, Lynn S.; Szilagyi, Peter G., “Bates' Guide to Physical Examination and History Taking”, 13e, Wolters Kluwer. 2021. Chapter 23: The Musculoskeletal System Subheadings: “Neck Pain” and “Low Back Pain”
- Casazza BA. Diagnosis and treatment of acute low back pain. Am Fam Physician. 2012 Feb 15;85(4):343-50. PMID: 22335313.
- Thompson, Jon C. *Netter's Concise Orthopedic Anatomy*, 2nd Edition. 2010 (Chapter 2: “Spine”).
- Stockard, Alan. *Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine*; Chapters 1, 5, 6
- Desai, Gautam. “Psoas Syndrome: A Frequently Missed Diagnosis. The Journal of the American Osteopathic Association, August 2012, Vol. 112, 522-528.
- DeJong's The Neurologic Examination, 7th Edition, William W. Campbell 2013; Chapter 37 (Introduction to Reflexes), Chapter 39 (Cutaneous) Reflexes
- *Harrison's Principles of Internal Medicine*, 19th Edition, Dennis L. Kasper, MD, Stephen L. Hauser, MD, J. Larry Jameson, MD, PhD, Anthony S. Fauci, MD, Dan L. Longo, MD and Joseph Loscalzo, MD, PhD 2015; Chapter 437 (Neurological Disorders)

Lecture Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HCY448FWYXREL4R>

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